

Mixed inputs in two-user Gaussian Interference Channels: Performance Analysis and Approximate Optimality

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Abstract

Necessary machinery for analyzing discrete inputs, or more generally mixed inputs (i.e. superposition of discrete and Gaussian random variables), is developed for multi user systems. The framework is rooted in a novel connection between information theory, additive combinatorics and number theory. The main idea is to use developed tools to analyze sub-optimal discrete point-to-point codes in which the reduction in achievable rates is more than compensated by decreasing interference created at the other users.

One of the application of the result show that i.i.d. mixed inputs are gDoF optimal and approximately optimal for finite SNR and INR (i.e. with in a gap) for the symmetric two-user Gaussian interference channel (G-IC). The framework can be easily extended to show optimality or approximate optimality results for variety of channels such as block asynchronous G-IC and G-IC with oblivious partial codebook knowledge (i.e. where joint decoding is not allowed).

Useful Bounds

Next bound makes connection between information theory, additive combinatorics and number theory.

Theorem 1. ^[1]
 $I_d(X_D) := [H(X_D) - \text{gap}(d_{\min}(X_D))]^+$
 $\leq I(X_D; X_D + Z) \leq H(X_D)$

$$\text{gap}(d) \leq \frac{1}{2} \log \left(\frac{2\pi e}{12} \right) + \frac{1}{2} \log \left(1 + \frac{12}{d^2} \right)$$

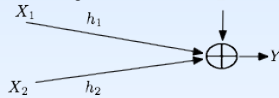
$$d_{\min}(X_D) = \min \{ |s_i - s_j| : s_i, s_j \in X_D, i \neq j \}$$

Remark: the bound is valid for any discrete input and any additive noise. For PAM we have that:

$$H(X_D) = \log(N), d_{\min}(X_D) = \sqrt{\frac{12\mathcal{E}_{X_D}}{N^2 - 1}}$$

Minimum Distance Bounds

In multi-user scenarios in order to apply the bound of the Theorem we must know cardinality and minimum distance of a resulting sum-set:



$$|h_1 X_{1D} + h_2 X_{2D}|$$

$$= \left\{ |h_1 x_{1D} + h_2 x_{2D}| : \begin{matrix} x_1 \in X_{1D}, \\ x_2 \in X_{2D} \end{matrix} \right\}$$

$$d_{\min}(h_1 X_{1D} + h_2 X_{2D})$$

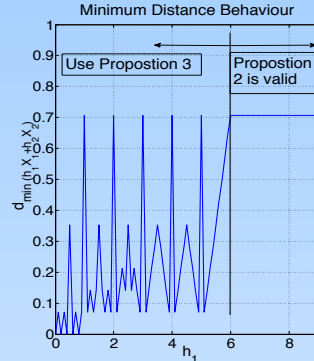
$$= \min \{ |s_i - s_j| : s_i, s_j \in h_1 X_{1D} + h_2 X_{2D}, i \neq j \}$$

Proposition 2. ^[2] Let $(h_x, h_y) \in \mathbb{R}^2$ be two constants. Let $X \sim \text{PAM}(|X|, d_{\min}(X))$ and $Y \sim \text{PAM}(|Y|, d_{\min}(Y))$. Then

$$|h_x X + h_y Y| = |X||Y|,$$

$$d_{\min}(h_x X + h_y Y) = \min \{ |h_x| d_{\min}(X), |h_y| d_{\min}(Y) \},$$

under the following conditions
either $|Y||h_y| d_{\min}(Y) \leq |h_x| d_{\min}(X)$
or $|X||h_x| d_{\min}(X) \leq |h_y| d_{\min}(Y)$.



When ever Proposition 2 does not apply we use.

Proposition 3. ^[1] Let $(h_x, h_y) \in \mathbb{R}^2$ be two constants. Let $X \sim \text{PAM}(|X|, d_{\min}(X))$ and $Y \sim \text{PAM}(|Y|, d_{\min}(Y))$. Then

$$|h_x X + h_y Y| = |X||Y| \quad (\text{a.e.})$$

and

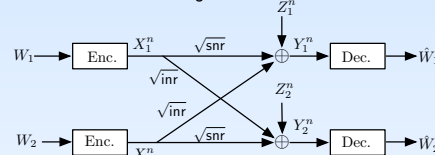
$$d_{\min}(h_x X + h_y Y) \geq \kappa_{\gamma, |X|, |Y|} \min \{ |h_x| d_{\min}(X), |h_y| d_{\min}(Y) \},$$

$$\max \left(\frac{|h_x| d_{\min}(X)}{|Y|}, \frac{|h_y| d_{\min}(Y)}{|X|} \right)$$

for all $(h_x, h_y) \in E \subseteq \mathbb{R}^2$ where $m(E^c) \leq \gamma$ for any $\gamma > 0$ and where $\kappa_{\gamma, |X|, |Y|} = \frac{\gamma}{2(1 + \ln(\max\{|X|, |Y|\}))}$.

Gaussian Channel (G-IC)

Channel model of G-IC is given shown below



Multi-letter Capacity Region

Multi-letter capacity expression for information stable IC is given by

$$\mathcal{C} = \lim_{n \rightarrow \infty} \text{co} \left(\bigcup_{P_{X_1^n X_2^n} = P_{X_1^n} P_{X_2^n}} \left\{ \begin{matrix} R_1 \leq \frac{1}{n} I(X_1^n; Y_1^n) \\ R_2 \leq \frac{1}{n} I(X_2^n; Y_2^n) \end{matrix} \right\} \right)$$

•Uncomputable

•Maximizing distribution is unknown

• Not clear whether there exists an equivalent single-letter form in general (there is for Strong Interference regime).

Single-letter Inner Bound Region

A simple inner bound to this expression is attained by using i.i.d. inputs and given by

$$\mathcal{R}_{\text{in}}^{\text{TIN}} = \bigcup_{P_{X_1 X_2} = P_{X_1} P_{X_2}} \left\{ \begin{matrix} R_1 \leq I(X_1; Y_1) \\ R_2 \leq I(X_2; Y_2) \end{matrix} \right\},$$

the above inner bound is commonly referred to as treating interference as noise (TIN) with out time sharing.

Useful metrics are:

the generalized Degrees of Freedom (gDoF) region

$$\mathcal{D}(\alpha) := \left\{ (d_1, d_2) \in \mathbb{R}_+^2 : \begin{matrix} d_i := \lim_{\substack{\text{snr} \rightarrow \infty \\ \text{snr} \rightarrow \infty}} \frac{R_i}{\frac{1}{2} \log(1 + \text{snr})}, \\ (R_1, R_2) \text{ is achievable} \end{matrix} \right\}.$$

and the capacity to within a gap

$$\mathcal{R}_{\text{in}} \subseteq \mathcal{C} \subseteq \mathcal{R}_{\text{Outer}} \subseteq \mathcal{R}_{\text{in}} - \text{gap},$$

where gap is:

•Bounded

•Constant (i.e. independent of channel parameters)

Main Results

Theorem 4: ^{[1],[2]}

For the G-IC the TIN-based achievable region $\mathcal{R}_{\text{in}}^{\text{TIN}}$ together with mixed inputs is gDoF optimal a.s. (i.e. up to a set of measure 0).

Interpretation: treating interference as noise without time sharing is gDoF optimal.

Theorem 5:

For the symmetric G-IC the TIN-based achievable region $\mathcal{R}_{\text{in}}^{\text{TIN}}$ is with in a gap of capacity given by:

- Very Weak Interference $\text{snr} \geq \text{inr}(1 + \text{inr})$,
gap = 1/2 bit,

- Weak Interference Type 2 $\text{snr} < \text{inr}(1 + \text{inr})$,

$$\frac{1 + \text{snr}}{1 + \text{inr} + \frac{\text{snr}}{1 + \text{inr}}} > \frac{1 + \text{inr} + \frac{\text{snr}}{1 + \text{inr}}}{1 + \frac{\text{snr}}{1 + \text{inr}}},$$

$$\text{gap} = \frac{1}{2} \log \left(\frac{26\pi e}{3} \right) \approx 4.39 \text{ bit},$$

- Weak Interference Type 1 $\text{inr} \leq \text{snr}$,

$$\frac{1 + \text{snr}}{1 + \text{inr} + \frac{\text{snr}}{1 + \text{inr}}} \leq \frac{1 + \text{inr} + \frac{\text{snr}}{1 + \text{inr}}}{1 + \frac{\text{snr}}{1 + \text{inr}}},$$

$$\text{gap} = \frac{1}{2} \log \left[\frac{2\pi e}{3} \left(1 + \frac{72(1 + \ln(\text{snr}))^2}{\gamma^2} \right) \right] \text{ bit},$$

- Strong Interference $\text{snr} \leq \text{inr} \leq \text{snr}(1 + \text{snr})$,

$$\text{gap} = \frac{1}{2} \log \left[\frac{2\pi e}{3} \left(1 + \frac{4(1 + \ln(\text{snr}))^2}{\gamma^2} \right) \right] \text{ bit},$$

- Very Strong Interference $\text{inr} \geq \text{snr}(1 + \text{snr})$,
gap = $\frac{1}{2} \log \left(\frac{2\pi e}{3} \right) \approx 1.25 \text{ bit}$,

for any $\gamma \in (0, 1]$.

Proof: consider the following mixed input

$$X_i = \sqrt{1 - \delta_i} X_{iD} + \sqrt{\delta_i} X_{iG}, \quad i \in [1 : 2],$$

$$X_{iD} \sim \text{PAM} \left(N_i, \sqrt{\frac{12}{N_i^2 - 1}} \right),$$

$$X_{iG} \sim \mathcal{N}(x_{iG}; 0, 1),$$

Next, rewrite mutual information as

$$I(X_2; Y_2) = I(X_2; \sqrt{\text{snr}} X_1 + \sqrt{\text{inr}} X_2 + Z_G)$$

$$= \left[h \left(\frac{\sqrt{1 - \delta_1} \sqrt{\text{inr}} X_{1D} + \sqrt{1 - \delta_2} \sqrt{\text{snr}} X_{2D} + Z_G}{\sqrt{1 + \text{inr} \delta_1 + \text{inr} \delta_2}} \right) - h(Z_G) \right]$$

$$+ \left[h \left(\frac{\sqrt{1 - \delta_1}}{\sqrt{1 + \text{inr} \delta_1}} \sqrt{\text{inr}} X_{1D} + Z_G \right) - h(Z_G) \right]$$

$$\leq \min \left(\log(N_1), \frac{1}{2} \log \left(1 + \frac{\text{inr}(1 - \delta_1)}{1 + \text{inr} \delta_1} \right) \right)$$

$$+ \frac{1}{2} \log(1 + \text{inr} \delta_1 + \text{snr} \delta_2) - \frac{1}{2} \log(1 + \text{inr} \delta_1)$$

The rest follows by Theorem 1 and Proposition 2 and 3 and optimizing over number of points and power splits.

Other Applications

This framework can be extend to variety of other channels. For example:

•**Interference Channel with Partial Codebook knowledge.**

This model studies scenarios when receivers have knowledge of only a subset of codebooks. Our framework shows that lack of interfering codebook knowledge at one of the receivers can result in at most 3.5 bit loss in performance. See [3],[4] for more details.

•**Block Asynchronous Gaussian Interference Channel**

This model assumes that interfering links are block asynchronous with the receivers. Unfortunately, this channel is not information stable and capacity must be expressed in terms of information density instead of mutual information. However, TIN inner bound is still a valid inner bound. Therefore, result of the previous section carry over to this channel as well.

List of Relevant Publications

[1]. A. Dytso, N. Devroye, and D. Tuninetti, "i.i.d. Mixed Inputs and Treating Interference as Noise are gDoF optimal for Symmetric Gaussian Two-user Interference Channel" IEEE Intern. Symp. on Inf. Theory (ISIT), 2015, submitted.

[2]. A. Dytso, N. Devroye, and D. Tuninetti, "On Gaussian interference channels with mixed Gaussian and discrete inputs," IEEE Intern. Symp. on Inf. Theory (ISIT), 2014, June 2014, pp.261–265.

[3]. A. Dytso, D. Tuninetti, and N. Devroye, "On discrete alphabets for the two-user Gaussian interference channel with one receiver lacking knowledge of the interfering codebook," in Info. Theory and Appl. Workshop (ITA), 2014, Feb 2014, pp. 1–8.

[4]. A. Dytso, D. Tuninetti, and N. Devroye, "On the two-user interference channel with lack of knowledge of the interference codebook at one receiver," Information Theory, IEEE Transactions on, vol. PP, no. 99, pp. 1–1, 2015.